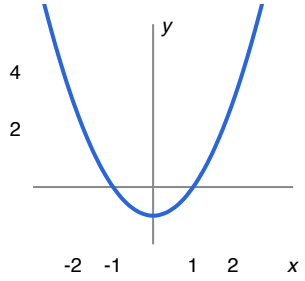
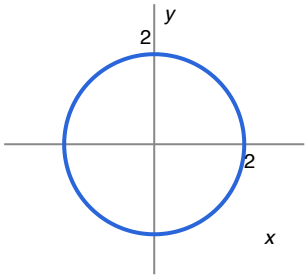
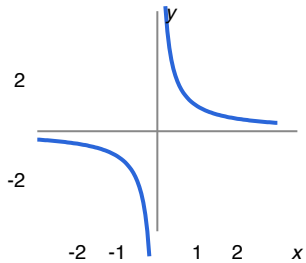
Algebra 8 — Functions · Domain and range · Composite and inverse functions

Marks in brackets. ★ = harder question. $\mathrm{fg}(x)$ means do g first, then f.

1 Decide, with a reason, whether each graph shows a function. [3]







a $y = x^2 - 1$ **b** the circle $x^2 + y^2 = 4$ **c** $y = \dfrac{1}{x}$

2 $\mathrm{f} : x \mapsto x^2 - 2x$. Find, simplifying your answers, [3]

a $\mathrm{f}(3x)$ **b** $3\mathrm{f}(x)$ **c** $\mathrm{f}(-x)$

3 The domain of $\mathrm{f} : x \mapsto x^2 + 1$ is all real numbers. Write down the range. [2]

4 State which values of x (if any) must be excluded from the domain of [3]

a $\mathrm{f}(x) = \sqrt{x + 2}$ **b** $\mathrm{f}(x) = x^3$ **c** $\mathrm{f}(x) = \dfrac{5}{2x - 1}$

5 $\mathrm{f}(x) = 3x + 1$ and $\mathrm{g}(x) = x - 4$. Find $\mathrm{fg}(5)$. [2]

6 $\mathrm{f}(x) = (x - 1)^2$ and $\mathrm{g}(x) = 3x$. Find [4]

a $\mathrm{fg}(x)$ **b** $\mathrm{gg}(x)$

7 Find the inverse function of [4]

a $\mathrm{f}(x) = \dfrac{x}{3} + 2$ **b** $\mathrm{f}(x) = \dfrac{4}{x+1}$

8 $\mathrm{f}(x) = 2(x-3)$ and $\mathrm{g}(x) = 2x + 5$. Given that $\mathrm{f}^{-1}(a) + \mathrm{g}^{-1}(a) = 4$, find a. [4]

Total: 25 marks

Check: function = one output for each input (vertical-line test) · $\mathrm{f}(2x)$: substitute $2x$ for every x; $2\mathrm{f}(x)$: double the whole thing · domain excludes division by 0 and square roots of negatives · $\mathrm{fg}(x)$ = g first · inverse: swap x and y, rearrange.